

OWASP Top 10 Research — Vortex Tech Cyber Security Internship (Week 1)

Project Overview

This repository contains my Week 1 submission for the Vortex Tech Cyber Security Internship Track. The task was to research five vulnerabilities from the OWASP Top 10 (2021) and document them in a way that explains what each one is, how it's exploited, a real-world case where it caused harm, and how developers can prevent it.

This is a research/documentation deliverable — no code or live testing was required this week. The goal was to build a solid theoretical foundation before moving into hands-on testing in later weeks.

Objectives

- Understand five OWASP Top 10 (2021) vulnerability categories in depth, beyond memorized definitions.
- Practice explaining technical security concepts in plain, accessible language.
- Study real, well-documented breaches to connect theory with actual impact.
- Build the habit of publishing internship work to a public GitHub portfolio.

Vulnerabilities Covered

Vulnerability	OWASP 2021 Category	Real-World Case Studied
SQL Injection	A03:2021 - Injection	TalkTalk (2015)
Broken Access Control	A01:2021 - Broken Access Control	Optus (2022)
Cryptographic Failures	A02:2021 - Cryptographic Failures	Adobe (2013)
Security Misconfiguration	A05:2021 - Security Misconfiguration	Capital One (2019)
Identification and Authentication Failures	A07:2021 - Auth Failures	23andMe (2023)

Full write-ups, including attack scenarios, prevention strategies, and severity ratings, are in [docs/OWASP-Top10-Research-Report.md](#).

Learning Outcomes

- Learned to distinguish between related-but-different concepts, like encryption vs. hashing, and authentication vs. authorization.
- Gained a clearer picture of how a single logic gap (e.g., a missing ownership check) can scale into a breach affecting millions of records.
- Practiced researching primary sources (regulator reports, official breach disclosures) rather than relying on secondhand summaries.
- Started thinking about security as an ongoing operational discipline (config audits, permission reviews) rather than a one-time code fix.

Repository Structure

```
vortextech-cybersec-week1/  
├── README.md           # This file  
├── docs/  
│   └── OWASP-Top10-Research-Report.md # Full research report (main deliverable)  
└── screenshots/        # (Reserved for future weeks' hands-on evidence)
```

Technologies / Resources Used

- OWASP Top 10:2021 — primary reference standard
- Public breach disclosures and regulator reports (ICO UK, OAIC Australia, DOJ US, company statements)
- Markdown for documentation formatting

Disclaimer

This repository is submitted as part of the Vortex Tech Cyber Security Internship Program (2026) and is for educational and portfolio purposes only. All real-world case studies referenced are based on publicly available information; no proprietary, confidential, or unpublished data is included. This week's task involved research and documentation only — no systems were accessed or tested.

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